

PHYSICAL CHEMISTRY I

CHEM 3610

Fall 2026

Instructor:	Matthew Rowley	Office Hours:	MTF 10:00 am – 11:00 am
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Please include the course number in the subject line of all correspondence.

Tentative Schedule

This class will meet on Mondays, Wednesdays, and Fridays from 12:00 to 12:50 pm in room 128 of the Science Center (SC).

For the best lecture experience, read the indicated textbook chapter *before* each lecture.

	Date	Topic	Chapter
Week 1	W, Sep. 2	The Perfect Gas and Kinetic Model	1A-1B
	F, Sep. 4	Real Gases	1C
Week 2	M, Sep. 7	Labor Day - No Class!	
	W, Sep. 9	Internal Energy	2A
	F, Sep. 11	Enthalpy	2B
Week 3	M, Sep. 14	Thermochemistry	2C
	W, Sep. 16	State Functions and Exact Differentials	2D
	F, Sep. 18	Adiabatic Changes	2E
Week 4	M, Sep. 21	Makeup/Review Day (Exam 1: Ch. 1–2)	
	W, Sep. 23	Entropy	3A
	F, Sep. 25	Entropy Changes	3B
Week 5	M, Sep. 28	The Measurement of Entropy and the System	3C-3D
	W, Sep. 30	Combining the First and Second Laws	3D
	F, Oct. 2	Phase Diagrams of Pure Substances	4A

	Date	Topic	Chapter
Week 6	M, Oct. 5	Thermodynamic Aspects of Phase Transitions	4B
	W, Oct. 7	Makeup/Review Day (Exam 2: Ch. 3–4)	
	F, Oct. 9	The Thermodynamic Description of Mixtures	5A
Week 7	M, Oct. 12	The Properties of Solutions	5B
	W, Oct. 14	Phase Diagrams of Binary Systems	5C-5D
	F, Oct. 16	Phase Diagrams of Ternary Systems	5E
Week 8	M, Oct. 19	Fall Break - No Class!	
	W, Oct. 21	Activities	5F
	F, Oct. 23	The Equilibrium Constant	6A
Week 9	M, Oct. 26	The Response of Equilibria to the Conditions	6B
	W, Oct. 28	Electrochemical Cells	6C-6D
	F, Oct. 30	Transport in Gases	16A
Week 10	M, Nov. 2	Motion in Liquids	16B
	W, Nov. 4	Diffusion	16C
	F, Nov. 6	Makeup/Review Day (Exam 3: Ch. 5, 6, 16)	
Week 11	M, Nov. 9	The Rates of Chemical Reactions	17A
	W, Nov. 11	Integrated Rate Laws	17B
	F, Nov. 13	Reactions Approaching Equilibrium	17C
Week 12	M, Nov. 16	The Arrhenius Equation	17D
	W, Nov. 18	Reaction Mechanisms	17E-17F
	F, Nov. 20	Photochemistry	17G
Week 13	M, Nov. 23	Thanksgiving Break - No Class!	
	W, Nov. 25	Thanksgiving Break - No Class!	
	F, Nov. 27	Thanksgiving Break - No Class!	

	Date	Topic	Chapter
Week 14	M, Nov. 30	Collision Theory	18A
	W, Dec. 2	Diffusion-Controlled Reactions	18B
	F, Dec. 4	Transition-State Theory	18C
Week 15	M, Dec. 7	The Dynamics of Molecular Collisions	18D
	W, Dec. 9	Electron Transfer in Homogeneous Systems	18E
	F, Dec. 11	Makeup/Review Day (Exam 4: Ch. 17–18)	
Finals Week	T, Dec. 15	Final Exam 11:00-12:50 Bring a pencil and a scantron sheet	